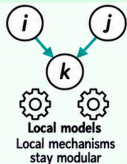
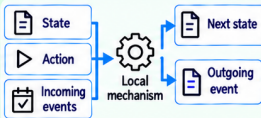


(a) Causal composition

Validated interfaces



Source module i



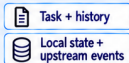
Destination module k



Composed rollout



(b) Optional local control



1 Upstream check

- ✓ Payment confirmed
- ✓ Inventory reserved

Shipment example

2 Predict effects



3 Gate action



4 Verify residual



Feedback for next round

Stop: task ends
or budget exhausted

Action sequence
+ outcome